

1 Evaluation and Entry of Serial Numbers

Serial numbers are assigned to be able to differentiate between individual items of material. Consequently, the serial number uniquely identifies a single part.

MES provides several variants with different features to record serial numbers:

- Variant 1: Entry of serial numbers for OPs that are not subject to management in batches (dialog A_SNR/ "E" only)
- Variant 2: Entry of serial numbers for OPs that are subject to management in batches (dialog A_SNR)
 - Manual entry of the serial number that is assigned to a HYDRA batch number.
 - Automatic assignment of the serial number = HYDRA batch number
 - Automatic assignment of the serial number that is assigned to a HYDAR batch number.

1.1 Entry of serial numbers for OPs that are not subject to management in batches (dialog A_SNR/ "E")

General / usage

Serial numbers are assigned in ERP when creating the production order. At the interface to HYDRA, assigned serial numbers are transferred as details for the order header and managed there.

The OP that is not subject to management in batches of the order is logged on and respective serial numbers are recorded in the relevant quality (yield/scrap) at AIP.

The serial number is entered manually (e.g. also by barcode on the single part) by the AIP user. To facilitate data collection, the serial numbers available for the order can also be displayed in a list. Only serial numbers may be logged on that have been assigned for this order. Please note that a serial number may exist as scrap only once.

The quantity recorded for the order/OP is always 1 for each single part and, depending on the quality, posted to the order's/OP's yield or scrap account.

Uploads to the ERP system in relation to the serial number (goods movements) can be performed for recorded serial numbers/single parts.

Configuration

The settings required to use the function for collecting serial numbers (without orders subject to management in batches) are described [here](#).

Usage at AIP

Basic AIP view

The user applies the "serial numbers" dialog (A_SNR). The relevant function key is configured in the basic view of AIP.

The screenshot displays the AIP (Advanced Production Interface) basic view. It features three main data tables and a bottom toolbar. The 'Machines/workplaces' table lists machines and their current status. The 'Operations on workplace' table shows production orders and their progress. The 'Produced output batches' table lists individual production batches. A red circle highlights the 'Collect serial number' button in the bottom toolbar, which is the function key for entering serial numbers.

Figure: Basic view with "serial number" function key (A_SNR)

Collection of serial numbers (A_SNR)

Serial number

The user enters the serial number. The serial number must be part of the order. A validation check is performed. If the serial number is not part of the order or if it is already recorded with a result, it cannot be used.

Clicking the function  opens a selection list showing the "free" serial numbers of the order.

Requesting the selection list simultaneously updates the list of already recorded serial numbers.

Grid

The displayed list shows all serial numbers in their relevant quality (yield/scrap) that have already been entered.

- Scrap is displayed in red.
- Yield is displayed in green.

Quality

When a serial number is entered, it has to be assigned a quality. You may choose from the following options:

- Yield
The serial number is entered with "yield" quality. The operation quantity is increased by yield = 1.
The default value is "yield".
- Scrap
The serial number is entered with "scrap" quality. The operation quantity is increased by scrap = 1.
- Scrap reason
If the "scrap" quality is entered, the user also has to enter or select a scrap reason.

Entry function

By clicking this function, the entered serial number is recorded with the selected quality.

List function

By clicking this function, the entire list of already entered serial numbers may be updated by the selected quality (e.g. when the dialog is reopened).

Interrupt/terminate OP for orders with serial numbers (A_UN/A_AB)

Quantities are not entered when operations requiring serial numbers are logged off. The quantity fields are to be disabled in these dialogs.

1.2 Entry of serial numbers for OPs that are subject to management in batches (dialog A_SNR)

General / usage

The following alternatives arise for entering and/or assigning serial numbers if an order/operation subject to batch management is used to enter serial numbers:

- Manual entry of the serial number that is assigned to a HYDRA batch number.
- Automatic assignment of the serial number = HYDRA batch number
- Automatic assignment of the serial number that is assigned to a HYDAR batch number.

Configuration

The settings required to use the function for collecting serial numbers (with orders subject to management in batches) are described [here](#).

Usage at AIP

Collection of serial numbers (A_SNR) including manual assignment of serial numbers - type "E"

An additional batch is created for each registered serial number in the area of material and production logistics if batch management requirement is enabled for the operation.

The relationship between the current output batch and the registered serial number is saved additionally in the database for traceability purposes. In this case, the output batch is considered as ID without inventory and, therefore, does not receive a quantity.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

The dialog for entering the serial number looks as described below. The serial number can be entered. Batches are assigned in the background.

Collection of serial numbers (A_SNR) including automatic assignment of serial numbers - type "G"

If the option "serial number requirement = G" is set for the operation, output batches will be recorded as serial numbers. In this case, the serial number is the output batch number.

For OPs subject to batch management, the dialog and relevant entries are as follows:



Collection of serial numbers (G)
<A_SNR>

Workplace
Operation

M20002500100

Material
SAL-FKA011-M-9050

Serial No.

PR4RH6B11H

Serial number	Class	Date	Time	*	Reason
PR4RH6B11G	A	04.07.2014	08:08:17	2	COLOR DEFECT
PR4RH6B11F	A	04.07.2014	08:08:09	1	BURR FORMATION
PR4RH6B11E	G	04.07.2014	08:07:46	0	
PR4RH6B11D	G	04.07.2014	08:07:45	0	

Quality

☐ Yield
☒ Scrap

Reason

0



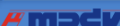
Staff Number

9999

Close

List




04.07.14 08:09:38

The "serial number" field is disabled and includes the current output batch. Once the badge number has been entered, posting can be performed by clicking "capture". If the entered quality is not "yield", a valid scrap reason has to be entered.

Exactly one batch with quantity 1 is created for each serial number and the serial number is the batch number.

The "list" function updates the list of already recorded serial numbers.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

Collection of serial numbers (A_SNR) including automatic assignment of serial numbers using the number range - type "S"

Serial numbers are recorded in relation to the batch if the option "serial number requirement = S" is set for the operation. In this case, the operation must be subject to management in batches. Every registered serial number causes an output batch to be changed (CA_WL) creating a batch with quantity 1 regarding the serial number in the MPL module.

The server determines a new serial number and displays it in the "serial number" field by clicking the function . The new serial number is assigned uniquely for the whole system using the "SNR" number range.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

Interrupt/terminate OP for orders with serial number tracking

Quantities are not entered when operations requiring serial numbers are logged off. The quantity fields are to be disabled in these dialogs.